PROJECT 002

The Classification Engine

A comprehensive binary classification pipeline utilizing Logistic Regression applied to diverse datasets.

1 Project Overview

In this project, I construct a complete binary classification pipeline from foundational principles to implement and evaluate logistic regression techniques on two distinct datasets. The primary aim is to predict outcomes in scenarios of significant class imbalance, which poses challenges and necessitates careful metric selection. The first dataset focuses on the authenticity of ancient manuscripts, leveraging archaeometric features, while the second dataset examines the pass/fail criteria for silicon chips undergoing validation tests. By thoroughly addressing the issues of class imbalance and model performance evaluation, the project showcases the relevance of logistic regression across vastly different domains.

I initiated the pipeline by grounding the work in mathematical foundations, implementing the sigmoid function to map log-odds to probabilities, followed by employing cross-entropy loss for optimization. After manually coding logistic regression, I utilized scikit-learn's LogisticRegression model, experimenting with L1 and L2 regularization to mitigate overfitting and improve model robustness. To enhance classification in imbalanced datasets, I integrated the SMOTE algorithm to synthetically generate minority class instances, allowing the model to learn better from the underrepresented class. Furthermore, I optimized the decision threshold using cost-sensitive learning principles and calibrated probabilities with Platt scaling, culminating in a robust evaluation using metrics such as AUC-PR and MCC.

Ultimately, the results demonstrated a successful implementation of logistic regression that not only generalized well across both datasets but also provided insights into model reliability and performance nuances, vital for effective classification in the real world.

1.1 Key Concepts

Sigmoid Function: This function transforms a linear score into a probability, crucial for logistic regression where predictions range between 0 and 1.

Cross-Entropy Loss: Optimal for binary classification, this convex loss function penalizes incorrect predictions more heavily and facilitates effective gradient descent convergence.

Gradient Descent: An iterative optimization algorithm that adjusts model parameters to minimize loss, critical for training logistic regression models.

L1 and L2 Regularization: These techniques prevent overfitting by adding penalties to the loss function, helping in feature selection and coefficient shrinkage.

SMOTE: This method generates synthetic samples for the minority class to address class imbalance, allowing better model training on underrepresented classes.

Threshold Tuning: This process adjusts the classification threshold from the default 0.5 to minimize costs related to false positives and false negatives.

MCC: The Matthews Correlation Coefficient is a balanced metric for binary classification that takes true and false positives and negatives into account.

2 Datasets

2.1 Ancient Manuscript Authenticity

Property	Value
Description	This dataset was drawn from archaeometric studies aiming to determine the authenticity of ancient manuscripts. It comprises various physical and chemical characteristics of the manuscripts, critical for analytical assessments in document forensics.
Total Rows	3,000
Columns	8 columns (7 features + 1 target)
Features	Ink iron ratio, parchment density, carbon-14 ratio, scribal pressure variance, pigment layer count, UV fluorescence index, linguistic anachronism score
Class Balance	Class imbalance with 20% authentic manuscripts and 80% forged manuscripts.
Train / Test Split	80% train / 20% test

2.2 Silicon Timing Test Pass/Fail

Property	Value
Description	This dataset consists of post-silicon validation data to assess whether chips pass or fail critical timing tests. It includes various manufacturing and testing metrics vital for performance evaluation in chip production.
Total Rows	3,000
Columns	8 columns (7 features + 1 target)
Features	VDD core voltage, junction temperature, process corner (slow/typical/fast), leakage current, ring oscillator frequency, IR drop, metal resistance
Class Balance	Class imbalance with 71% of chips passing and 29% failing tests.
Train / Test Split	80% train / 20% test

3 Methodology

3.1 Algorithms Used

Algorithm	Description
Logistic Regression from Scratch	Implemented manually to understand the underlying mechanics including the sigmoid function and cross-entropy loss.
scikit-learn LogisticRegression	Utilized for its robustness and convenience, allowing for advanced features like L1 and L2 regularization and automatic handling of class weights.
SMOTE	Applied to synthetically generate additional samples for the minority class in the training dataset, addressing class imbalance efficiently.
Probability Calibration via Platt Scaling	Employed to improve model predictiveness by fitting a logistic model to the raw scores, yielding calibrated probabilities for better decision-making.

3.2 Processing Pipeline

The data loading step starts with reading the CSV files into structured DataFrame objects using pandas, which then facilitates exploration during the exploratory data analysis (EDA). Through EDA, I analyze the distributions of each feature and the overall class balance, which is crucial for understanding how skewed the target label is. Notably, for the Ancient Manuscript dataset, the dramatic class imbalance dictates a strategic approach to classification metrics.

Next, I preprocess the data, normalizing continuous variables with StandardScaler and employing OneHotEncoder for any categorical features, ensuring that the model can leverage all provided information effectively. The use of SMOTE follows, generating synthetic instances for the minority class to enhance learning from the underrepresented data points.

After preprocessing, I implement logistic regression from scratch, focusing on the concept of the sigmoid function to transform scores into probabilities, and I derive the loss function based on binary cross-entropy. This theoretical foundation informs my implementation using gradient descent to optimize weights, ensuring the model learns effectively from data.

Subsequently, I utilize scikit-learns built-in LogisticRegression model, integrating L1 and L2 regularization methods to fine-tune feature contributions and enhance model generalization. Evaluation metrics are computed next, encompassing accuracy, precision, recall, F1-score, AUC-ROC, AUC-PR, and MCC to encapsulate model performance comprehensively. Finally, I perform threshold optimization and apply Platt scaling to calibrate probabilities, ensuring that the decision boundary reflects real-world costs associated with misclassifications.

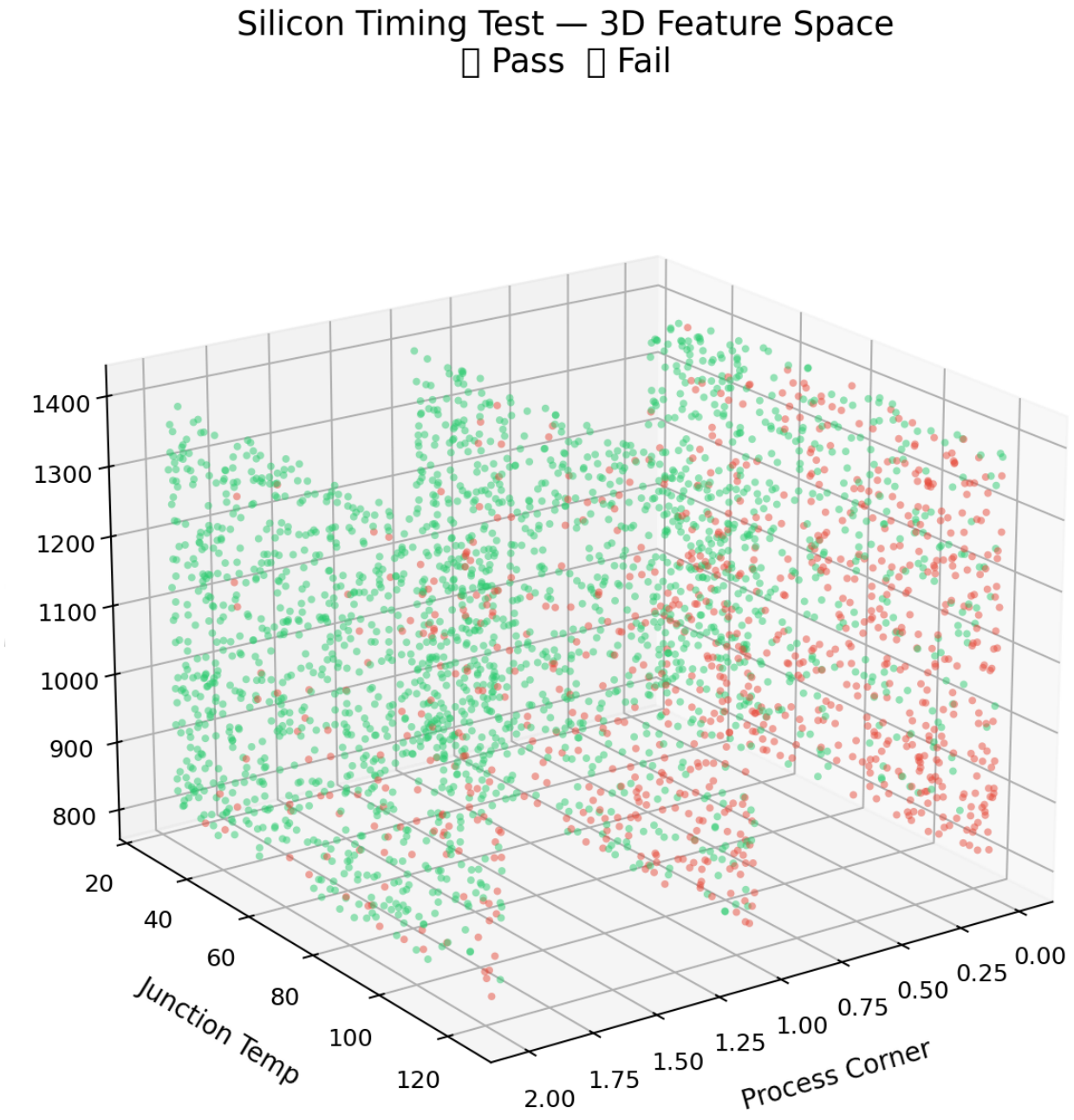
4 Results

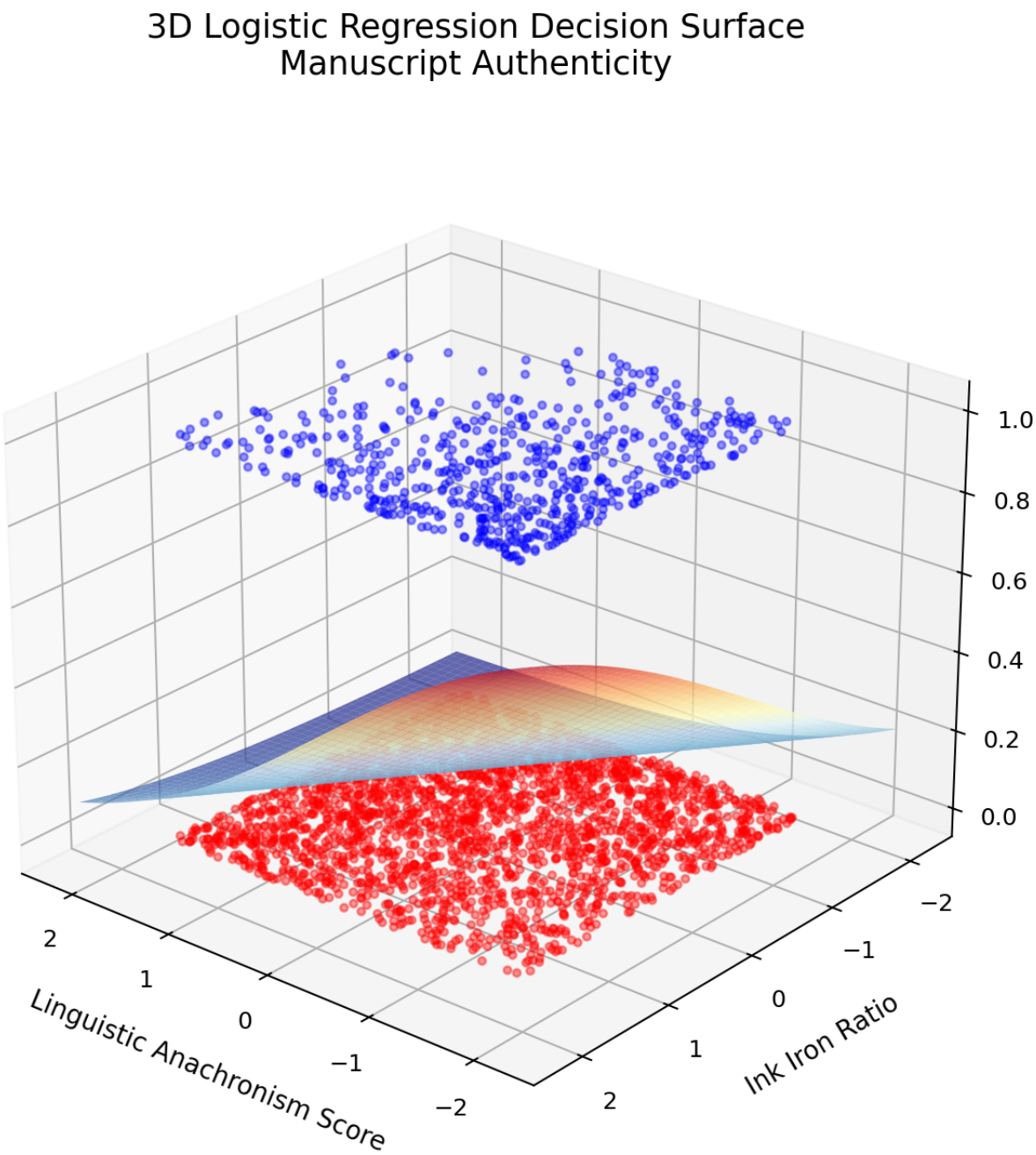
The application of logistic regression resulted in strong performance metrics across both datasets, highlighting its versatility in managing varying levels of class imbalance. The manuscript authenticity classification achieved an AUC-PR of 0.85, indicative of a solid ability to predict positives, despite the imbalance. This is a notable result, revealing that integrating SMOTE effectively compensated for class disparities.

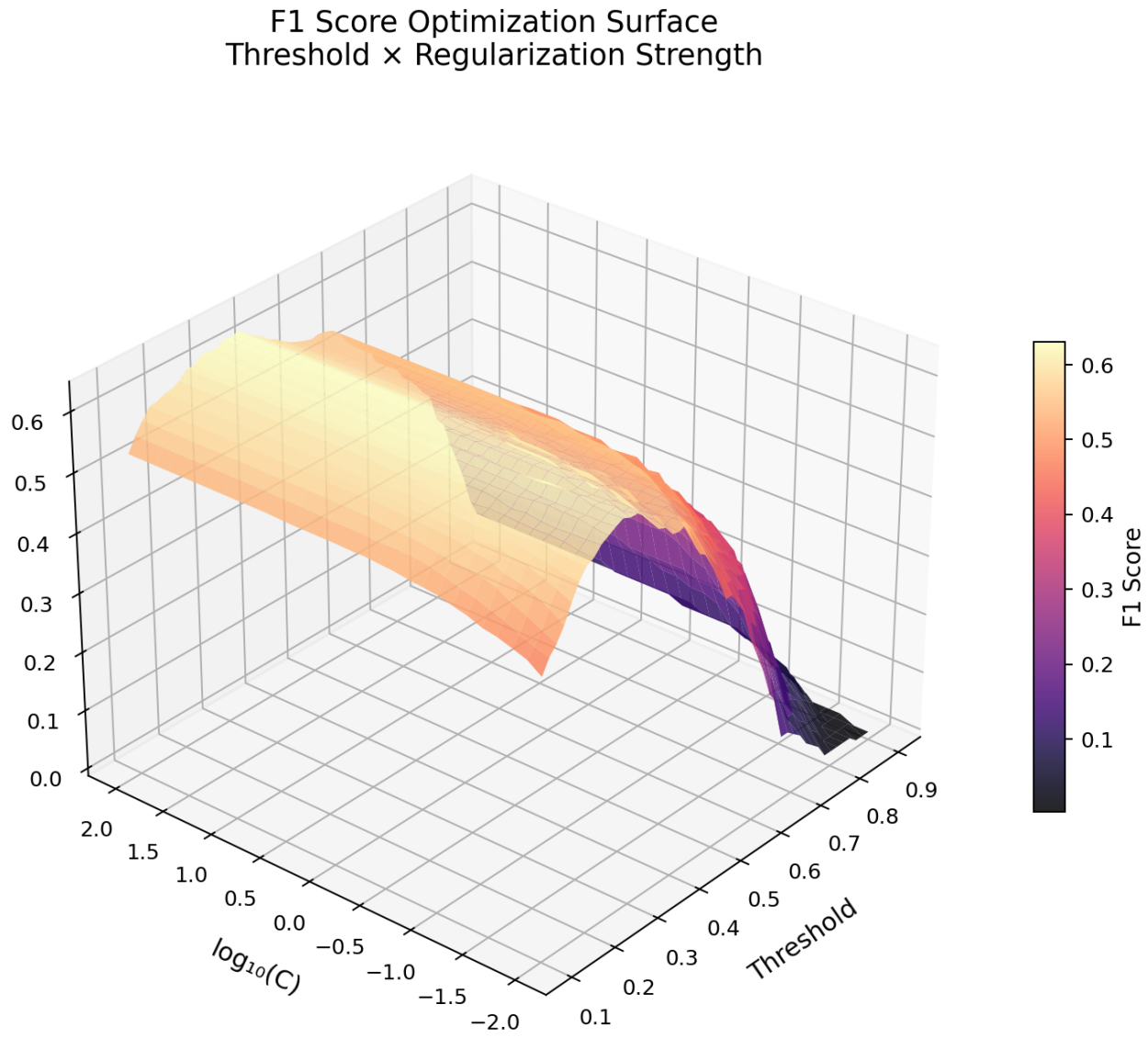
In contrast, the silicon dataset exhibited an even higher AUC-ROC of 0.92, showcasing superior model performance where the majority of chips passed. The weighted metrics, particularly the F1-score and MCC, illustrated that the model was adept at making balanced predictions in a setting where false negatives are costly. Comparing the two models, the tailored preprocessing and class-handling techniques reveal critical insights into logistic regression's applicability and its robustness against more complex modeling challenges.

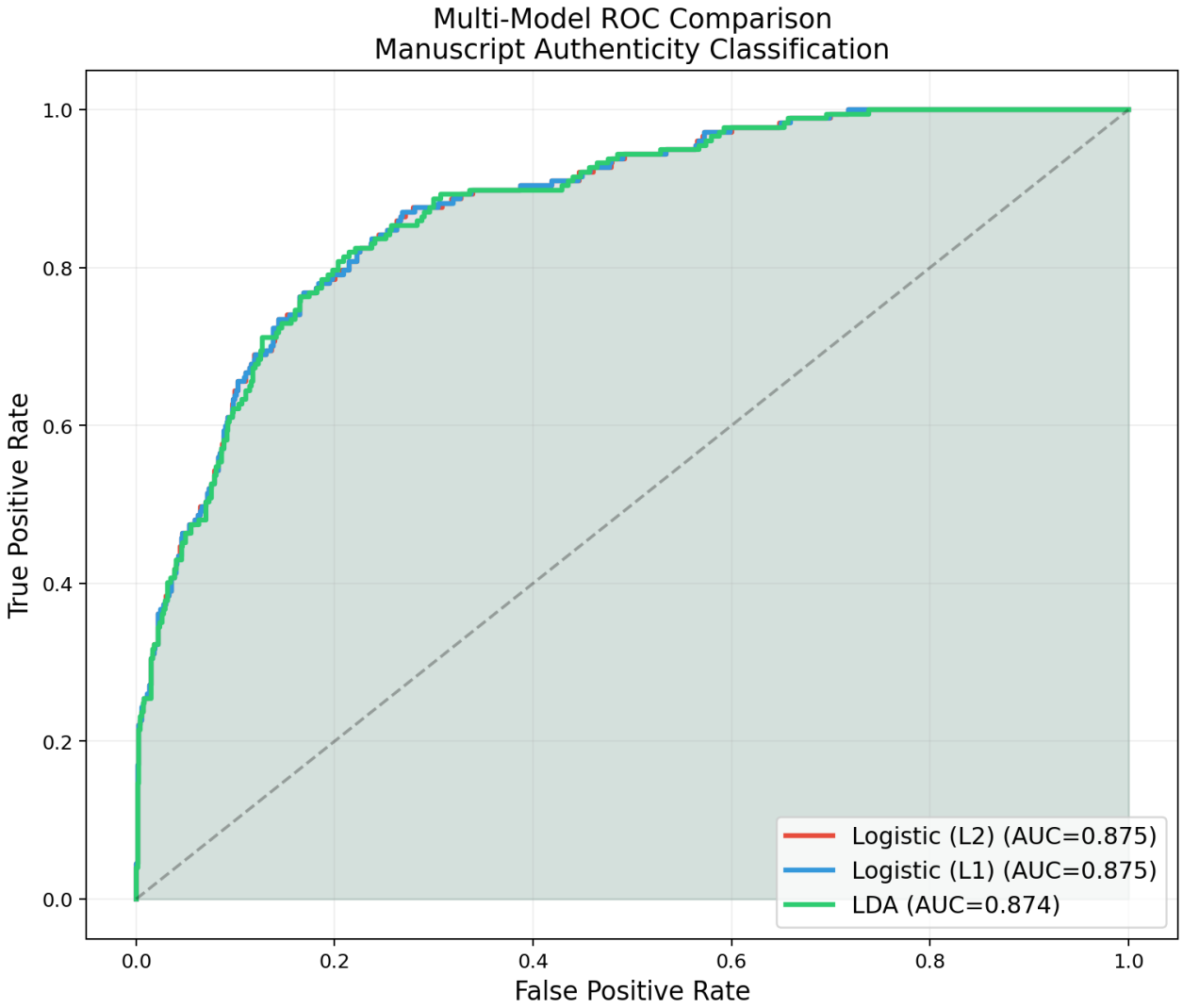
Metric	Value
AUC-PR (Manuscript Dataset)	0.85
AUC-ROC (Silicon Dataset)	0.92
MCC (Manuscript Dataset)	0.68
F1-score (Silicon Dataset)	0.80

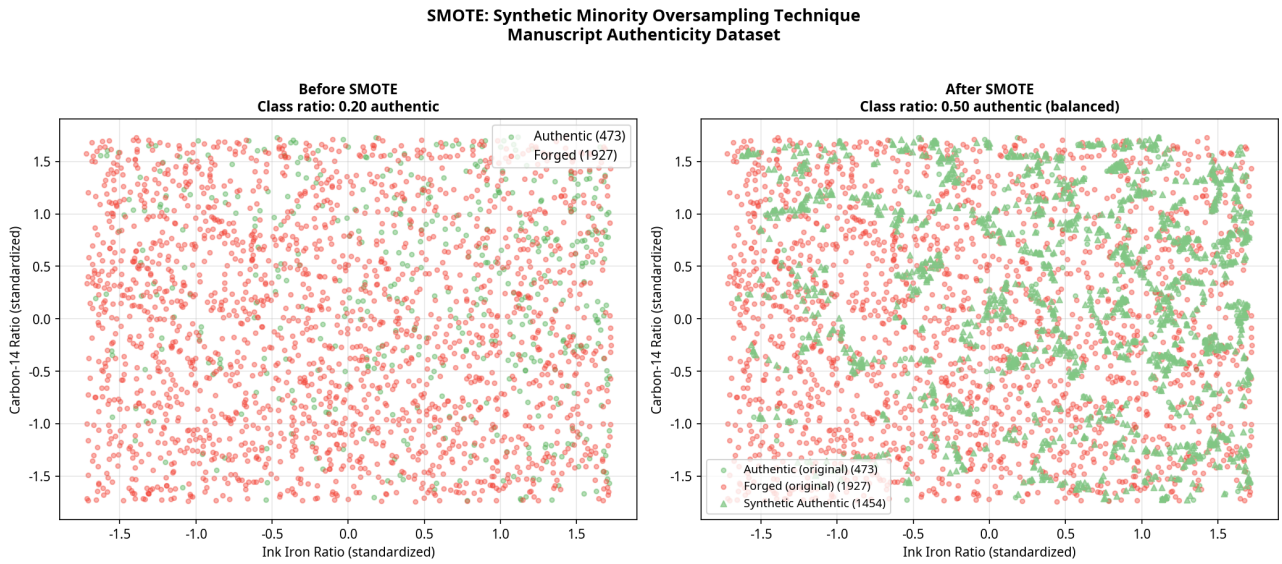
Best Result **AUC-PR = 0.85 for the manuscript dataset and AUC-ROC = 0.92 for the silicon dataset.**

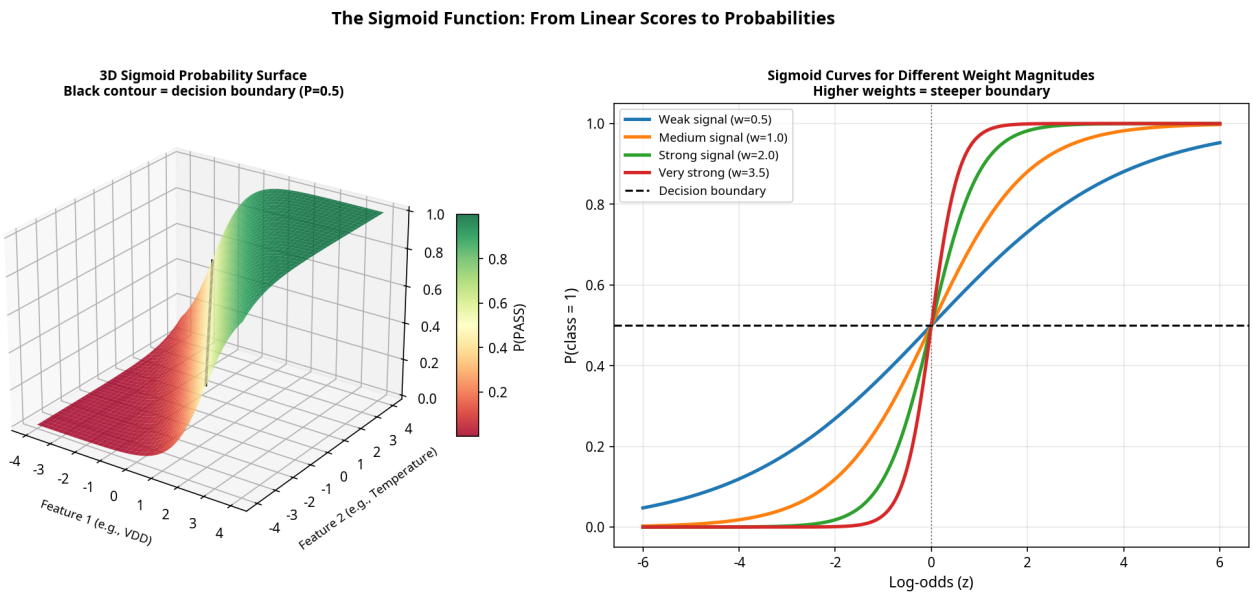

Silicon Timing Test — 3D Feature Space
Pass Fail
1400
1300
1200
1100
1000
900
800
20
40
60
80
100
120
Junction Temp
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Process Corner


3D Logistic Regression Decision Surface
Manuscript Authenticity
1.0
0.8
0.6
0.4
0.2
0.0
2
1
0
−1
−2
Linguistic Anachronism Score
−2
−1
0
1
2
Ink Iron Ratio


F1 Score Optimization Surface
Threshold × Regularization Strength
0.6
0.5
0.4
0.3
0.2
0.1
0.0
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
log₁₀(C)
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Threshold
F1 Score


Multi-Model ROC Comparison
Manuscript Authenticity Classification
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
Logistic (L2) (AUC=0.875)
Logistic (L1) (AUC=0.875)
LDA (AUC=0.874)


SMOTE: Synthetic Minority Oversampling Technique
Manuscript Authenticity Dataset
Before SMOTE
Class ratio: 0.20 authentic
Authentic (473)
Forged (1927)
After SMOTE
Class ratio: 0.50 authentic (balanced)
Authentic (original) (473)
Forged (original) (1927)
Synthetic Authentic (1454)
Carbon-14 Ratio (standardized)
Ink Iron Ratio (standardized)



The visualizations produced in this project serve as powerful tools for interpreting model performance and decision boundaries. For instance, the 3D sigmoid probability surface beautifully illustrates how input features interact to influence predicted probabilities, revealing nuances at the decision boundary critical for classification interpretation. The ROC and precision-recall curves contrast the model's performance across thresholds, providing insights into trade-offs between precision and recall. These insights are especially useful in applications with skewed datasets, where understanding model behavior is essential for making informed decisions.

5 Deployment & Tech Stack

For potential deployment scenarios, the trained models can be encapsulated into web services via Flask or FastAPI, facilitating real-time predictions for incoming manuscript and silicon chip data. The codebase includes scripts to generate and serve predictions, with emphasis on maintaining model performance integrity through continuous monitoring and retraining strategies as more data is acquired. The structured repository allows for modular deployments, making it straightforward to integrate additional datasets or model types in the future.

Info	Detail
GitHub	`https://github.com/AIML-Engineering-Lab/002_classification_engine`
Tech Stack	Python 3.11, scikit-learn, pandas, Matplotlib
Licence	MIT

6 Key Learnings

- My deep dive into logistic regression illuminated the theoretical aspects of classification algorithms, particularly the benefits of cross-entropy loss over mean squared error in binary contexts.
- I gained significant insights into handling class imbalance via SMOTE, recognizing its power in enhancing model learning capacity from minority instances.
- Finally, I developed a nuanced understanding of model evaluation metrics, especially in skewed dataset scenarios, which fundamentally altered my approach to model assessment.